SkillVersed — Data Analytics with Gen AI

Excel — Phase 1, Level 2

UK Data Dashboard Project — Project Report

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Dataset: DfT Road Safety Data (STATS19) — Collisions, 2021–2025

1. Dataset Selection

The dataset selected for this project is the UK Department for Transport's Road Safety Data (STATS19), specifically the Collisions table covering the last five reporting years. Transport was chosen from the brief's list of suitable domains because road safety data combines a clear time dimension, rich categorical detail (road type, weather, lighting, speed limit, police force area), and genuine public-interest relevance — collision patterns directly inform infrastructure and enforcement decisions.

Several UK-domain datasets were compared before this one was chosen — NHS waiting-time data (healthcare), police recorded crime (crime and policing), and school performance tables (education) were all considered. STATS19 Collisions was selected because it natively satisfies a strict data-quality bar (a minimum of 1,000 rows, at least three genuine numeric measures, and at least three category fields, using only real, non-synthetic data) more cleanly than the alternatives: several other candidates offered only one or two native numeric fields and would have needed derived measures to qualify.

The data was sourced directly from gov.uk/DfT rather than a third-party mirror, which is the primary source STATS19 mirrors on other platforms are themselves copied from.

2. Dataset Evaluation and Documentation

2.1 Description

Source	Department for Transport — Road Safety Data (STATS19), gov.uk
Table used	Collisions (last 5 reporting years)
Row count	513,801
Column count	43
Geographic scope	Great Britain, collision-level records with police force area, local authority and lat/long detail
Granularity	One row per collision
Time period	5 most recent reporting years available from DfT at time of download

2.2 Reasons for selection

Beyond meeting the numeric/category requirements natively, this dataset was chosen because it is the exact worked example the brief itself uses to illustrate a strong analytical perspective (road type and time-of-day driving casualty patterns), and because collision-level granularity supports genuine multi-angle analysis — by road type, time, speed limit, and force area — within a single table, avoiding the added complexity of a multi-table join within an 8-day window.

2.3 Strengths

- Official government source with a documented, versioned coding scheme.
- Large sample size (513,801 rows) supports reliable aggregation without small-sample distortion, provided volume guards are applied when ranking by rate.
- Rich category set (road type, weather, lighting, severity, urban/rural, police force area, day of week, hour) supports genuine multi-perspective analysis.
- Comes with an official DfT data guide, enabling verifiable decoding of every categorical field.

2.4 Limitations

- Categorical fields are stored as integer codes, not readable labels, and required decoding via the official 2024_code_list lookup table before analysis was possible.
- Several fields use “-1” as a missing/unknown-value code (e.g. junction_detail_historic, carriageway_hazards_historic) rather than a blank cell; these needed explicit handling during cleaning.
- The coding scheme changed partway through the historical record, leaving duplicate-purpose column pairs (e.g. junction_detail_historic vs junction_detail; carriageway_hazards_historic vs carriageway_hazards). The current-scheme column was kept and the historic equivalent dropped from analysis to avoid redundant or conflicting fields.
- January 2021 shows an artificially low collision count (5,570 vs a typical 9,000+) due to COVID-19 lockdown restrictions rather than a genuine road-safety improvement. The dashboard's monthly trend chart starts from 2022 to avoid this misleading the reader; the 2021 dip is documented here rather than presented as a trend.
- A small number of collisions (n=3) record an “Unknown” speed limit; these were excluded from the speed-limit vs severity chart as they are too few to support a reliable proportion.

3. End-Consumer Identification

3.1 Potential stakeholder groups

- Local authority road safety / transport planners deciding where to prioritise engineering or enforcement interventions
- Police force area commanders reviewing resourcing against collision patterns
- Insurers assessing regional and road-type risk

- Road safety campaign groups and journalists investigating regional disparities
- Central government (DfT) monitoring national trend against road safety targets

3.2 Primary audience

The dashboard is designed for a road safety / transport planner deciding where to prioritise intervention — for example a local authority or National Highways planner allocating a limited engineering or enforcement budget across road types, locations and times.

3.3 Justification

This audience was chosen because every finding the dataset surfaces — road type, speed limit, time of day, day of week, and police force area — maps directly onto levers this role actually controls or influences: where to redesign a junction, where to set speed enforcement, and when to schedule patrols. A vaguer audience such as “the general public” would give no basis for deciding what to cut from the dashboard, whereas naming this specific decision-maker made every layout and chart choice defensible against a single test: does this help a planner prioritise faster?

4. Perspective and Analysis Definition

4.1 Perspectives considered

- An overview of all UK road collisions by year — rejected as too broad; doesn't name a decision or a decision-maker.
- A focus on casualty types (pedestrians vs vehicle occupants) — rejected because the collisions table itself doesn't carry casualty-level detail; that lives in a separate STATS19 table not included in this project's scope.
- A focus purely on speed-limit vs severity — rejected as too narrow; it answers only one of several questions a planner needs and would under-use the dataset's category richness.
- Where and when serious and fatal collisions concentrate, and what road/speed/time conditions drive that concentration — selected.

4.2 Selected perspective

“Which road types, speed limits and times of day/week account for the highest concentration of serious and fatal collisions across Great Britain (2021–2025), so a road safety planner can prioritise where to intervene?”

4.3 Rationale

This question names a measure (serious/fatal collision rate), a comparison (across road type, speed limit, time and force area), a period (2021–2025, with the 2021 lockdown anomaly explicitly handled), and a decision (where to prioritise intervention). It matters because serious and fatal collisions — not the much larger volume of slight-severity collisions — are what road safety interventions are ultimately judged

against, and because the answer is actionable: it names conditions (60mph rural single carriageway, late afternoon) rather than describing the data in the abstract.

5. Dashboard Design and Development

5.1 Layout decisions

The dashboard fits a single screen with no scrolling, as required. A navy title bar runs across the top, carrying the dashboard's finding-led title, the named audience, and the date range. Below it, five KPI cards give the headline numbers — total collisions, serious or fatal, serious/fatal rate, worst hour, and single-carriageway share — with amber top-borders on the two severity/rate cards and teal on the volume cards, so the KPI row already establishes the colour logic the charts continue. The slicer panel is pinned in a vertical column on the left so filters are visible rather than hidden, and is limited to the three dimensions that actually drive a chart on this dashboard (year, urban/rural, police force area) rather than every available category field, keeping the panel functional rather than decorative. The six visuals sit in a 2×3 grid to the right of the slicer panel, ordered left-to-right and top-to-bottom in the order a planner would naturally ask the questions: overall trend and road type first, then the peak-hour and speed/severity relationship, then the day/time pattern and the force-area ranking last, since that ranking is best read once the other patterns are understood.

5.2 Visualisation choices

- Line chart — serious/fatal collisions vs all collisions per month, 2021–2025. A line is the standard, honest choice for a continuous time series, and keeps the January 2021 lockdown dip visible and labelled rather than hidden.
- Horizontal bar — serious/fatal collisions by road type, sorted descending. Bar is the correct form for ranked category comparison; horizontal avoids cramped labels for longer category names.
- Column chart — serious/fatal collisions by hour of day, restricted to the six highest hours (13:00–18:00) so the peak window reads clearly. Column suits a natural left-to-right time axis better than a horizontal bar would.
- 100% stacked column — severity composition (fatal/serious/slight) at each speed limit. Stacking to 100% is the honest way to show a changing proportion across a category, rather than raw counts which would be dominated by road volume rather than risk.
- 100% stacked area — time-of-day mix across four representative days (Monday, Friday, Saturday, Sunday). This chart was originally built as a line, but a line implies continuity between adjacent points, and the four days are non-consecutive categories rather than a true sequence — a slope between “Monday” and “Friday” is not a real trajectory. Converting it to a stacked area keeps the same finding (weekend nights roughly double a weekday night, while the AM peak nearly halves) without that misleading implication, and gives the dashboard a fifth distinct chart type.
- Bar chart — top police force areas by serious/fatal rate, restricted to areas with at least 1,000 collisions (a value filter on the underlying PivotTable) so the ranking reflects genuine risk rather than a small force's result being distorted by a low sample size.

In total the dashboard uses five distinct chart types — line, bar, column, 100% stacked column, and 100% stacked area — comfortably clearing the brief's minimum of four, with each choice defensible on the grounds given above rather than chosen for variety's own sake.

5.3 User experience considerations

Colour is used consistently throughout on a three-colour ramp — navy for lower-emphasis values, light teal for mid-range values, and amber reserved exclusively for the highest or most severe value in each chart (the 17:00 peak-hour bar, the fatal-severity band, Police Scotland's rate, the single-carriageway share). Because amber consistently means “the one that matters most,” the eye is trained to find it the same way on every chart.

To keep that meaning intact when slicers are used, colour was assigned by each bar's position (rank) rather than by its category name or a fixed colour per category. Since every relevant PivotTable is already sorted (by value or by a fixed axis), position and rank coincide: whichever category climbs into the top slot after a slicer change automatically inherits amber, without needing a macro. This was a deliberate choice over VBA, since a macro-driven solution would require saving as .xlsm, and the brief requires a final .xlsx deliverable — macros are stripped on save, which would silently break the colour scheme for the marker. The trade-off, noted here rather than hidden, is that position-based colouring assumes a stable number of categories; if a slicer changes how many bars a chart displays, the banding should be re-checked.

Every chart title states the finding it shows rather than a generic subject line (for example “Single carriageways carry 78% of serious/fatal collisions” rather than “Collisions by road type”), and the peak-hour column chart's colouring was specifically checked against its title — the amber highlight sits on 17:00 alone, so the visual emphasis and the stated finding agree. KPI numbers use thousands separators and consistent hour formatting (17:00) so they read at a glance without mental conversion, and the dashboard title carries an explicit audience line — “for road safety planners prioritising interventions” — so the intended reader is stated up front rather than left implicit.

5.4 Benefits of this design approach

For a road safety planner, the dashboard answers “where should I look first” within seconds via the KPI row and top-left chart, then supports progressively narrower questions (which hour, which speed limit, which force area) without ever requiring a scroll or a second screen. Because every chart is driven from a shared PivotTable cache, the year/area/urban-rural slicers let the planner re-run the same five findings for a specific region or year without rebuilding anything — directly supporting the kind of prioritisation decision this audience actually makes.

6. Workbook Structure

The workbook uses a five-sheet structure. The brief's own illustrative four-sheet diagram (raw → clean → analysis → dashboard) does not on its own account for the fact that STATS19's categorical fields are

stored as coded integers rather than readable labels. A fifth sheet, holding the official DfT 2024_code_list as a live reference table, was therefore added between the raw and clean stages — this is what every decoding formula in the clean-data sheet looks up against, keeping the decoding traceable to a real, citable source rather than hardcoded.

Sheet	Contents
01_Raw_Data	The DfT collision CSV exactly as downloaded, 513,801 rows x 43 columns. Left untouched throughout the project as an audit trail.
02_Code_List	The official DfT 2024_code_list reference table, imported as a live sheet. Every decoding formula in 03_Clean_Data looks up against this sheet rather than hardcoded values.
03_Clean_Data	A cleaned copy of the raw data: categorical fields decoded from integer codes to readable labels via lookups against 02_Code_List, missing-value codes handled, and redundant historic-scheme columns removed.
04_Analysis	Seven PivotTables (monthly trend, road type, hour of day, speed x severity, police force area, day x time band, KPI summary), all built from one shared PivotCache so a single slicer set drives every table.
05_Dashboard	The finished single-screen dashboard: KPI cards, six charts, and the slicer panel. Contains no raw data, only visuals and formula-driven KPI cells.

6.1 Cleaning process

Calculation mode was set to Manual before any decode formula was added, since each new column recalculates across all 513,801 rows. The full sequence carried out on Clean Data was as follows.

Duplicate check on collision_index

A COUNTIF-based approach was rejected as computationally infeasible on a 513,801-row column (it would require roughly 264 billion comparisons). Instead, total row count was compared against the count of distinct values:

- **Total rows —** `=COUNTA(dft_road_casualty_statistics_collision_last_5_years3[collision_index])`
- **Distinct values —** `=ROWS(UNIQUE(dft_road_casualty_statistics_collision_last_5_years3[collision_index]))`

The two figures were equal, confirming collision_index is a genuine unique key with zero duplicate rows.

Date and time format check

Dates were displaying in US order (m/d/yyyy) on a UK dataset. Before reformatting, COUNT was used to confirm every entry in both fields was a genuine number rather than text masquerading as a date:

- **Date validity —** `=COUNT(dft_road_casualty_statistics_collision_last_5_years3[date])`
- **Time validity —** `=COUNT(dft_road_casualty_statistics_collision_last_5_years3[time])`

Both returned 513,801, confirming no text entries. The date column was then reformatted to Custom dd/mm/yyyy and the time column to Custom hh:mm — a display-only change, since the underlying serial values were already correct.

Decoding categorical fields

Each categorical field was decoded from its DfT integer code into a readable label using the official 2024_code_list lookup table, with SWITCH used for short, known code sets and XLOOKUP used for the one long list (police force area, 52 codes):

- **Severity** — `=SWITCH([@[collision_severity]],1,"Fatal",2,"Serious",3,"Slight","Unknown")`
- **Is_Serious_Fatal (rate flag)** — `=IF([@collision_severity]<=2,1,0)`
- **Road_Type** — `=SWITCH([@road_type],1,"Roundabout",2,"One way street",3,"Dual carriageway",6,"Single carriageway",7,"Slip road",12,"One way/Slip road","Unknown")`
- **Road_Class** — `=SWITCH([@first_road_class],1,"Motorway",2,"A(M)",3,"A",4,"B",5,"C",6,"Unclassified","Unknown")`
- **Speed_Limit_MPH** — `=IF(OR([@speed_limit]=-1,[@speed_limit]=99),"Unknown",[@speed_limit]&" mph")`
- **Light_Conditions** — `=SWITCH([@light_conditions],1,"Daylight",4,"Dark - lights lit",5,"Dark - lights unlit",6,"Dark - no lighting",7,"Dark - lighting unknown","Unknown")`
- **Urban_Rural** — `=SWITCH([@urban_or_rural_area],1,"Urban",2,"Rural",3,"Unallocated","Unknown")`
- **Day_Name** — `=SWITCH([@day_of_week],1,"Sunday",2,"Monday",3,"Tuesday",4,"Wednesday",5,"Thursday",6,"Friday",7,"Saturday","Unknown")`
- **Hour** — `=HOUR([@time])`
- **Time_Band** — `=IFS([@Hour]<6,"00-06 Night",[@Hour]<10,"06-10 AM peak",[@Hour]<16,"10-16 Midday",[@Hour]<19,"16-19 PM peak",TRUE,"19-24 Evening")`
- **Month_Start** — `=DATE(YEAR([@date]),MONTH([@date]),1)` — formatted mmm yyyy
- **Police_Force_Area** — `=XLOOKUP([@police_force],'2024_code_list'!$C$9:$C$60,'2024_code_list'!$D$9:$D$60,"Unknown")`

Decode verification

Every decoded field was checked across all 513,801 rows (not just a sample) by counting how many rows fell into the “Unknown” catch-all:

- **Example (Severity)** — `=COUNTIF(dft_road_casualty_statistics_collision_last_5_years3[Severity],"Unknown")`

Severity, Day_Name, Urban_Rural and Police_Force_Area returned zero unknowns, confirming complete decoding on fields with no missing source data. Road_Type, Road_Class, Speed_Limit_MPH and Light_Conditions returned a small non-zero count, which is expected: the DfT -1 (“data missing or out of range”) sentinel deliberately falls into each formula's Unknown catch-all rather than being silently

dropped. One check initially returned a much larger figure (13,060 for Road_Type) than the missing-data audit implied — investigation showed the check formula was pointing at the raw numeric column rather than the decoded column, so it was comparing numbers against the text “Unknown” and could never match. Once corrected to reference the decoded field directly, Road_Type also returned zero unknowns. This is recorded here as a worked example of verifying a check formula itself, not just the data it checks.

Final unknown counts across all decoded fields: Severity 0, Road_Type 0, Road_Class 6, Speed_Limit_MPH 3, Light_Conditions 34, Urban_Rural 8, Day_Name 0, Police_Force_Area 0 — 51 unknown values out of 513,801 rows (0.01%), concentrated entirely in fields affected by the DfT sentinel convention rather than indicating a flaw in the decoding logic.

Freezing to values

Once all twelve decoded columns were verified, they were converted from live formulas to static values (select columns → copy → Paste Special → Values) before any PivotTable was built. With roughly 6.2 million formula cells across the twelve columns, this step is what keeps PivotTables and slicers responsive; the mapping remains fully traceable via the formulas documented above and the 2024_code_list reference sheet.

7. Key Insights

The five findings below are what the dashboard's KPI row and six charts surface, and what directly informed the choice of primary audience and perspective in Sections 3 and 4.

Finding	Detail
Road type	Single carriageways carry 77.8% of all serious and fatal collisions (96,767 of 124,366).
Time of day	17:00 is the single worst hour (10,363 serious/fatal collisions); 15:00–18:00 forms a clear PM peak block.
Speed limit	Fatal share rises steadily with speed limit: 0.47% at 20mph, 0.87% at 30mph, 1.80% at 40mph, 2.89% at 50mph, 3.84% at 60mph. Serious-or-fatal share is 34.5% on 60mph roads vs 22.6% on 30mph roads.
Day of week	Friday is the worst day overall (20,278), Sunday the lowest (15,686) — but weekend nights (Fri/Sat) run roughly double a weekday night.
Rurality	Rural roads account for 39.8% of serious/fatal collisions despite carrying far less traffic than urban roads; rural single carriageways alone make up 31% of all serious/fatal collisions in Great Britain.

8. Dashboard Screenshot

Figure 1 — finished dashboard, “Where and when serious collisions concentrate,” built for a road safety planner prioritising intervention (Great Britain, 2021–2025).

Where and When Serious Collisions Concentrate

For road safety planners prioritising intervensions

2021-2025

Slicers

2021 | 2022 | 2023 | 2024 | 2025

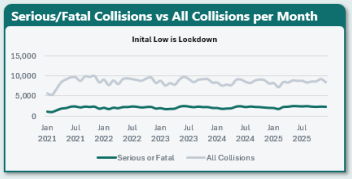
Rural | Unallocated | Unknown | Urban

Dark - lighting unknown | Dark - lights lit | Dark - lights unlit | Dark - no lighting | Daylight | Unknown

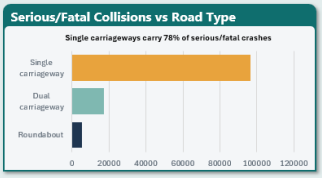
A | A(M) | B | C | Motorway | Unclassified | Unknown

Total Collisions	Serious or Fatal	Serious/Fatal Rate	Worst Hour	Total Fatalities
513801	124366	24.21%	17	7553

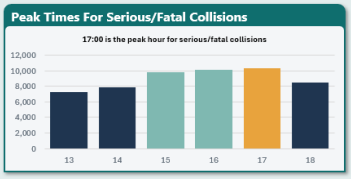
Serious/Fatal Collisions vs All Collisions per Month



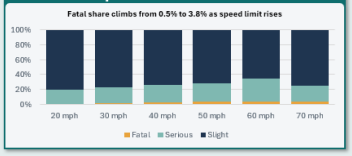
Serious/Fatal Collisions vs Road Type



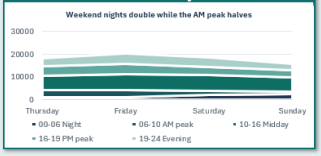
Peak Times For Serious/Fatal Collisions



Collision % vs Speed



Serious/Fatal Collisions vs Days



Top Serious/Fatal Collisions Rate Per Police Force

